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Is Generative Artificial Intelligence Ready for Computational Social Science?*

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Abstract. This study analyse the potential of Large Language Models (LLMs) for social simulation by evaluating their ability to: (a) align with principles of economic rationality; (b) make decisions consistent with explicit preferences; (c) exhibit socially rational behaviour; and (d) refine beliefs to anticipate the actions of other agents. Using a series of game-theoretic experiments, this paper assess the performance of models such as `GPT-4.5`, `Mistral-Small`, and `Qwen3`. The results indicate that while some models behave consistently in simple settings, they struggle in more complex scenarios that require strategic anticipation. This study highlights both the promise and the current limitations of LLM-based generative agents in computational social science.

Keywords: Generative Autonomous Agents and Multi-Agent Systems, Large Language Models, Social Simulation, Game theory

1 Introduction

Generative Autonomous Agents and Multi-Agent Systems (GAAMAS), powered by Large Language Models (LLMs), have recently demonstrated remarkable capabilities in simulating mathematical reasoning, decision-making, and human behaviours [7, 9, 13, 21, 24]. These advancements have sparked growing interest in their application to computational social sciences, where researchers aim to model, analyse, and predict social dynamics using agent-based modelling [8, 12, 18]. The potential of LLM-powered agents lies in their ability to generate human-like responses, adapt to complex environments, and interact with multiple agents. However, their reliability, consistency, and capacity for high-level reasoning remain open questions [4, 5, 23].

This paper explores the potential of *homo silicus* for social simulation by examining the behaviour exhibited by intelligent machines [20], with particular attention to whether generative agents deviate from established principles of economic and social rationality in ways similar to humans. Drawing on behavioral game theory, the study assesses the capabilities of models such as `GPT-4.5` [17], `Llama3` [14], `Mistral-Small` [15],

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`DeepSeek-R1` [3], and `Qwen3` [19] to make coherent one-shot decisions, generate algorithmic strategies based on explicit preferences, adhere to first- and second-order rationality principles, and update beliefs in response to the behaviours of other agents[1]. The contributions of this work are as follows:

- An analysis of the capacity of generative agents to align with economic and social principles.
- An examination of their rationality, highlighting both strengths and limitations in decision-making contexts.
- An assessment of their ability to refine beliefs through interaction and to anticipate the decisions of other agents.
- A discussion of current limitations of generative GAAMAS for social simulation and proposed directions for future improvement.

Through this analysis, the study aims to assess whether GAAMAS can serve as reliable tools for computational social science research and to identify the advancements necessary to enhance their ability to credibly simulate human behaviour.

Following the review of related work in Section 2, I examine the behaviour of generative agents with respect to economic rationality, and their ability to act in accordance with explicit preferences in Section 3. Section 4 evaluates their capacity for social rationality, while Section 5 investigates how they refine beliefs and anticipate the actions of other agents. The paper concludes in Section 6, where I summarize the main contributions and propose directions for future research.

2 Related Work

A Multi-Agent System (MAS) consists of multiple autonomous computing entities, called agents, that interact within an environment and participate in an organization. As an explanatory model of social dynamics, a MAS is characterized by the emergence of complex behaviours that cannot be captured through predictive or solvable equations.

A central challenge in designing MAS lies in the fact that the internal mechanisms governing social behaviours are not directly observable. Agent architectures typically rely on predefined rules (e.g., finite state machines), symbolic equations (e.g., social force models), stochastic models (e.g., SIR), or reinforcement learning algorithms (e.g., *Q-learning*). These approaches depend on either knowledge engineering or trial-and-error learning to approximate realistic or optimal behaviours. By contrast, LLMs, trained on massive textual corpora, are capable of zero-shot or few-shot learning [22, 24]. This allows them to produce more nuanced and realistic representations of decision-making processes, communication, and adaptation. LLMs have been shown to emulate human behaviour in certain experimental economics and social psychology settings [13], and exhibit exceptional performance in zero-shot scenarios [9, 24]. LLM-powered agents can simulate perception, reasoning, and decision-making; adaptively respond to dynamic environments without explicit instructions; autonomously generate goals; and communicate using natural language [7, 21].

[1] All code, prompts, and data traces are available on a pubic repository [16]

Park *et al.* investigate the emergence of social behaviours in a generative agent-based simulation inspired by the video game *The Sims* [18], implemented in the open-source project *AI Town*[2]. The believability of LLM-driven agent behaviours and the resulting social dynamics depend on an architecture that enables agents to retrieve relevant events, reflect on their memories, and generate coherent plans and reactions.

Liu *et al.* propose a stable alignment framework based on simulated social interactions using three stages: imitation, self-criticism, and realignment [12]. Social norms emerge as generative agents engage in discussions on controversial societal topics or risk-related issues. Gao *et al.* introduce a generative MAS named `S3` (*Social-network Simulation System*) to simulate content propagation in social networks [8]. Their framework models user-generated content while accounting for demographic characteristics, contextual information, and the cognitive mechanisms underpinning decision-making. In this system, LLMs are used to infer a user's likelihood to share or create content based on incoming information, allowing simulations to reflect complex interactions between user profiles and social dynamics.

Despite these promising advances in modeling emergent collective behaviours with GAAMAS, several fundamental questions remain. For example, what types of social science problems are best suited to this approach? Under what conditions is agent behaviour credible? In pursuit of answering such questions, several preliminary studies have evaluated the game-theoretic capabilities of LLMs.

Chen *et al.* assess the economic rationality of LLM-generated decisions across various domains [2]. Their findings suggest that LLMs often make more rational choices than humans, though their performance is highly sensitive to context. In contrast, our work focuses on evaluating the intrinsic economic rationality of generative agents in environments designed to be as neutral and controlled as possible.

Fontana *et al.* study the cooperative behaviour of generative agents in the iterated prisoner's dilemma against opponents with varying hostility levels [5]. They propose a method to evaluate, *ex post*, whether agents understand the game rules and consider interaction history. However, such understanding is not necessarily integrated into the decision-making process. Our study instead emphasizes evaluating whether agents explicitly incorporate beliefs and opponent modelling into their strategies.

Willis *et al.* explore cooperative behaviour in generative MAS using natural language to guide LLMs in constructing fixed strategies for the iterated prisoner's dilemma, which are then translated into Python algorithms [23]. Although their work explores different behavioural attitudes (e.g., aggressive, cooperative, neutral), it does not verify whether these strategies truly reflect the intended attitudes or rational decision-making. In contrast, we focus on whether agents' behaviours integrate explicit preferences, beliefs about others, and knowledge of the game structure—whether in fixed strategies or one-shot decisions.

Fan *et al.* highlight the limitations of LLMs in game-theoretic reasoning [4]. They conclude that existing models are incapable of simulating rational decision-making. However, their study is not reproducible: the code is unavailable, and the experiments are limited to proprietary LLMs, which required significant computational resources and incurred substantial carbon costs. In contrast, we make our code, prompts, and

[2] https://github.com/a16z-infra/ai-town

datasets openly available [16], and we focus on open-weight models that can run on standard hardware. Furthermore, we minimize environmental impact by encouraging LLMs to generate algorithmic strategies rather than issuing multiple one-shot queries.

3 Economic reasoning

Decision-making plays a central role in social simulation, as it enables the modeling of individual agent behaviours within complex systems. It is a cognitive process that results in the selection of a course of action from among several alternatives, based on the agent's preferences.

To evaluate the economic rationality of different LLMs, I introduce an investment game that tests whether the models follow stable decision-making patterns or react inconsistently to changes in the game's parameters (see Section 3.1).

To explore the underlying preferences of generative agents, I use the dictator game — a simplified version of the ultimatum game. In this game, a single player (the dictator) decides how to allocate an endowment (e.g., a monetary amount) between themselves and a recipient, who has no influence on the outcome. Section 3.2 evaluates the decisions made by LLMs acting as the dictator and interpret their choices as a reflection of their intrinsic preferences. Section 3.3 provides specific instructions that encode preferences, allowing to assess the models' ability to incorporate these instructions into their decision-making.

3.1 Economic rationality

An investor allocates a basket $x_t = (x_t^A, x_t^B)$ of 100 points between two assets: Asset A and Asset B. The value of these points depends on randomly generated prices $p_t = (p_t^A, p_t^B)$, which determine the monetary return per allocated point. For example, if $p_t^A = 0.8$ and $p_t^B = 0.5$, each point assigned to Asset A yields \$0.80, while each point allocated to Asset B yields \$0.50. The game is repeated over 25 rounds to evaluate the consistency of the investor's choices.

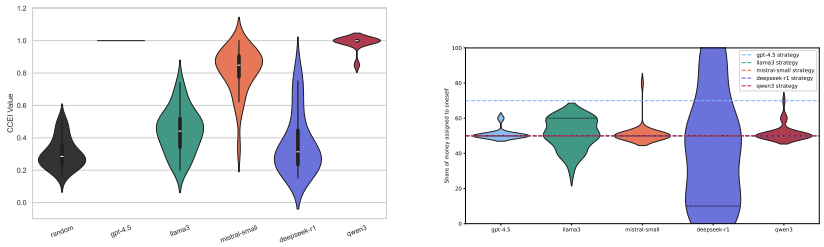
To assess the rationality of these decisions, we employ Afriat's Critical Cost Efficiency Index (CCEI) [1]. The CCEI is a standard metric in experimental economics used to evaluate adherence to the Generalized Axiom of Revealed Preference (GARP), a core criterion for rational decision-making. When choices deviate from consistency, the CCEI quantifies the minimal proportional adjustment to the budget needed to make the observed decisions consistent with utility maximization. The budget for each round is computed as

$$I_t = p_t^A \cdot x_t^A + p_t^B \cdot x_t^B.$$

The CCEI is obtained by solving a linear optimization problem that identifies the largest $\lambda \in [0,1]$ such that all observed decisions satisfy the adjusted constraint $p_t \cdot x_t \leq \lambda I_t$. This indicates how much the budget must be reduced to render the choices consistent with rationality. Human decision-makers typically achieve high CCEI scores—often close to 1, whereas lower values reflect irrational or inconsistent behaviour.

To ensure deterministic outputs and evaluate consistency, each model completes 30 iterations of the game with a fixed temperature of 0.0. As shown in Fig. 1a, the models

display notable variation in their consistency. `GPT-4.5` achieves a perfect CCEI score of 1.0, indicating flawless rationality. `Qwen3` and `Mistral-Small` follow closely, exhibiting stable and high CCEI values. `Llama3` performs moderately, with scores ranging from 0.2 to 0.74. In contrast, `DeepSeek-R1` displays considerable inconsistency, with CCEI values ranging from 0.15 to 0.83.



(a) CCEI Distribution per model

(b) Dictator's share per model

Fig. 1: Assessing economic and preference alignment

3.2 Preference Elicitation

To investigate the underlying preferences of generative agents, I interpret the decisions made by a LLM. Each model is prompted to produce both a one-shot decision and a corresponding strategy for the dictator game, implemented in `Python`. All one-shot experiments are repeated 30 times with a fixed temperature of 0.7.

Figure 1b shows a violin plot of the amount (out of $100) each model allocates to itself. Across `GPT-4.5`, `Llama3`, `Mistral-Small`, `DeepSeek-R1`, and `Qwen3`, the median share under one-shot conditions is $50—likely influenced by corpus-level biases such as term frequency. When asked to generate a strategy, all models opt for equal division, except `GPT-4.5`, which retains approximately 70% of the total. Notably, human participants under similar conditions typically keep around $80 on average [6]. When the LLMs are assigned the role of a human rather than an assistant, only `Llama3` deviates, with a median share of $60. Unlike algorithmically defined strategies, the variability in one-shot actions across samples can serve to simulate human behavioural diversity influenced by context, experience, or individual preferences.

Sensitivity analysis confirms that the retained portion remains stable across temperatures. However, lower temperatures lead to more deterministic outputs, while higher temperatures increase allocation diversity, suggesting more random exploratory behaviour.

3.3 Preference Alignment

Four preferences are defined for the dictator, each reflecting a specific notion of social welfare: (1) Egoism maximizes the dictator's income; (2) Altruism maximizes the re-

cipient's income; (3) Utilitarianism maximizes the total income; and (4) Egalitarianism maximizes the minimum income between the players.

Four allocation options are presented, where part of the money is lost during the division process, each corresponding to one of the defined preferences:

- Dictator: \$500, Recipient: \$100, Loss: \$400 (Egoistic);
- Dictator: \$100, Recipient: \$500, Loss: \$400 (Altruistic);
- Dictator: \$400, Recipient: \$300, Loss: \$300 (Utilitarian);
- Dictator: \$325, Recipient: \$325, Loss: \$350 (Egalitarian).

Table 1 assesses the models' ability to align with these preferences. When generating strategies, all models align perfectly except `DeepSeek-R1`, which fails to produce valid code, and `Qwen3`, which does not follow egoistic or altruistic strategies but adheres to utilitarian and egalitarian preferences. For one-shot action generation, `GPT-4.5` exhibits strong alignment overall but shows reduced adherence to utilitarian preferences. `Llama3` aligns well with egoistic and altruistic options but performs less consistently on utilitarian and egalitarian ones. `Mistral-Small` demonstrates high alignment with altruism, moderate alignment with utilitarianism, and low accuracy for egoistic and egalitarian choices. `DeepSeek-R1` aligns primarily with utilitarianism, showing weak alignment elsewhere. `Qwen3` strongly follows utilitarian preferences and moderately aligns with altruism but fails to exhibit egoistic behavior and shows limited adherence to egalitarianism.

Model	**Egoistic**	**Altruistic**	**Utilitarian**	**Egalitarian**
Strategy				
`GPT-4.5`	1.00	1.00	1.00	1.00
`Llama3`	1.00	1.00	1.00	1.00
`Mistral-Small`	1.00	1.00	1.00	1.00
`DeepSeek-R1`	–	–	–	–
`Qwen3`	0.00	0.00	1.00	1.00
Action				
`GPT-4.5`	1.00	1.00	0.50	1.00
`Llama3`	1.00	0.90	0.40	0.73
`Mistral-Small`	0.40	0.93	0.76	0.16
`DeepSeek-R1`	0.06	0.20	0.76	0.03
`Qwen3`	0.00	0.80	0.93	0.36

Table 1: Accuracy of dictator decisions based on preferences and models

Errors in action selection may arise from arithmetic miscalculations (e.g., incorrectly assuming that $\$500 + \$100 > \$400 + \$300$) or from misinterpretations of the intended preferences. For instance, `DeepSeek-R1`, when selecting a utilitarian option, justifies its choice with the explanation, "I think fairness is key here" indicating confusion between utilitarianism and egalitarianism.

In summary, most of LLMs generally align well with stated preferences but encounter greater difficulty when generating individual actions compared to algorithmic

strategies. DeepSeek-R1 fails to produce valid strategies and shows weak alignment in action selection.

4 Strategic reasoning

An autonomous agent is considered rational in a social interaction if it selects the optimal action based on its beliefs about the situation. The agent satisfies second-order rationality if it is not only rational itself but also believes that other agents are rational. In other words, a second-order rational agent chooses actions by both optimizing its own outcome and anticipating the rational decisions of others. Experimental game theory studies indicate that 93% of human subjects demonstrate rationality, while 71% exhibit second-order rationality [10].

To assess the first- and second-order rationality of generative autonomous agents, a simplified version of the ring-network game is used. This setup involves two players, each with two possible actions, aiming to maximize their own payoff. The payoff matrix corresponds to Variant A in Table 2.

Version	**a**		**b**		**c**		**d**	
Player 1 \Player 2 (version)	**A**	**B**	**A**	**B**	**A**	**B**	**A**	**B**
X	15	5	8	7	6	5	15	5
Y	0	10	7	8	0	10	0	40

Table 2: Four variants of the payoff matrix of the ring-network game

If Player 2 is rational, they must choose A, as B is strictly dominated. Player 1, if rational, may choose either X or Y: X is the best response if they believe Player 2 will choose A, while Y is optimal if they expect Player 2 to choose B. However, if Player 1 satisfies second-order rationality, they must choose X, anticipating that Player 2 will act rationally. To mitigate potential naming biases in LLMs, the action labels are reversed in half of the experiments.

Three types of beliefs are considered: an *implicit belief*, where the optimal action must be inferred from the natural language description of the payoff matrix; an *explicit belief*, where it is clearly stated that action B is strictly dominated by A; and a *given belief*, in which the optimal choice for Player 1 is directly provided in the prompt. The evaluation first examines the agents' rationality, followed by their second-order rationality.

4.1 First-Order Rationality

Table 3 evaluates the models' ability to generate rational behavior for Player 2. When generating strategies, GPT-4.5 and Mistral-Small consistently exhibit rational behavior, while Llama3 follows a random strategy. DeepSeek-R1 fails to produce valid outputs and Qwen3 displays irrational choices. In action generation, GPT-4.5 reliably

selects rational decisions, even under implicit belief conditions. Mistral-Small outperforms other open-weight models in this task. Both Llama3 and Qwen3 struggle to identify optimal actions based solely on implicit information. DeepSeek-R1 proves unreliable for simulating rational agents.

Model	Given	Explicit	Implicit
Strategy			
GPT-4.5	1.00	1.00	1.00
Mistral-Small	1.00	1.00	1.00
Llama3	0.50	0.50	0.50
Deepseek-R1	–	–	–
Qwen3	0.00	0.00	0.00
Actions			
GPT-4.5	1.00	1.00	1.00
Mistral-Small	1.00	1.00	0.87
Llama3	1.00	0.90	0.17
Deepseek-R1	0.83	0.57	0.60
Qwen3	1.00	0.93	0.50

Table 3: Accuracy of Player 2's decisions under different belief types

4.2 Second-Order Rationality

To adjust the difficulty of optimal decision-making, we define four variants of the payoff matrix for Player 1 in Table 2: (a) the original configuration, (b) a reduced gap between the payoffs, (c) an increased payoff for the suboptimal choice Y, and (d) a decreased payoff for the optimal choice X.

Table 4 evaluates the models' ability to generate second-order rational behavior for Player 1.When generating strategies, GPT-4.5 consistently exhibits second-order rational behavior in all configurations except (b), where it fails to distinguish the optimal action from a nearly optimal one. Llama3 makes decisions randomly, showing no strong pattern of rational behavior. In contrast, Mistral-Small demonstrates strong capabilities across all conditions, consistently generating second-order rational behavior. Qwen3 generates irrational strategies, and DeepSeek-R1 does not produce valid responses in strategy generation.

When generating actions, GPT-4.5 performs well in the initial configuration (a) but encounters significant difficulties when the payoff structure changes. Llama3 performs adequately under explicit and given belief conditions but struggles with implicit beliefs, especially as the payoff matrix becomes more complex. Mistral-Small performs well with given or explicit beliefs but shows weaknesses under implicit belief conditions, particularly in configuration (d). Meanwhile, DeepSeek-R1 struggles with lower accuracy overall, especially under implicit beliefs. Qwen3 performs robustly across most

belief types, particularly in configurations (a) and (b), maintaining strong scores under both explicit and implicit conditions. However, like other models, it experiences a noticeable drop in accuracy under implicit beliefs in configuration (d), suggesting sensitivity to deeper inferential reasoning.

Model	**a**			**b**			**c**			**d**		
	Given	**Explicit**	**Implicit**	**Given**	**Explicit**	**Implicit**	**Given**	**Explicit**	**Implicit**	**Given**	**Explicit**	**Implicit**
Strategy												
`GPT-4.5`	1.00	1.00	1.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	1.00
`Llama3`	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
`Mistral-Small`	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
`Deepseek-R1`	-	-	-	-	-	-	-	-	-	-	-	-
`Qwen3`	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Actions												
`GPT-4.5`	1.00	1.00	1.00	1.00	0.67	0.00	0.86	0.83	0.00	0.50	0.90	0.00
`Llama3`	0.97	1.00	1.00	0.77	0.80	0.60	0.97	0.90	0.93	0.83	0.90	0.60
`Mistral-Small`	0.93	0.97	1.00	0.87	0.77	0.60	0.77	0.60	0.70	0.73	0.57	0.37
`Deepseek-R1`	0.80	0.53	0.56	0.67	0.60	0.53	0.67	0.63	0.47	0.70	0.50	0.57
`Qwen3`	1.00	1.00	1.00	0.90	0.96	1.00	1.00	0.96	0.70	1.00	0.96	0.46

Table 4: Accuracy of Player 1's decisions in the different variants

Irrational decisions are explained by inference errors based on the natural language description of the payoff matrix. For example, in variant (d), the `Mistral-Small` model with given beliefs justifies its poor decision as follows: "Since player 2 is rational and A strictly dominates B, player 2 will choose A. Given this, if I choose X, I get fewer points (15) than if I choose Y (40). Therefore, choosing Y maximizes my gain."

In summary, `GPT-4.5` demonstrates the strongest performance in both first- and second-order rationality, though it occasionally struggles with distinguishing between optimal and near-optimal decisions. `Mistral-Small` also performs well but encounters difficulties with implicit beliefs, particularly in second-order reasoning. In contrast, `DeepSeek-R1` and `Llama3` exhibit inconsistent and often irrational decision-making, frequently failing to generate valid strategies. `Qwen3` similarly struggles to produce valid strategies, reflecting limited capabilities in high-level planning.

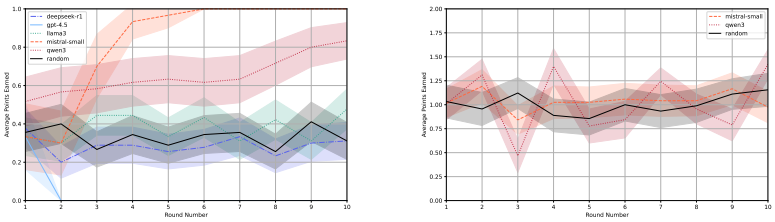
5 Epistemic reasoning

As discussed in the previous section, beliefs—whether implicit, explicit, or given—are essential to an autonomous agent's decision-making process. They enable the agent to anticipate the actions of others.

To evaluate the ability to refine beliefs in predicting an interlocutor's next move, a simplified version of the Rock-Paper-Scissors (RPS) game is used. In this setup, the opponent follows a hidden strategy (a repetition model), and the agent must predict the opponent's next move (Rock, Paper, or Scissors). A correct prediction earns 1 point, while an incorrect one earns 0. The game is played over N rounds, and the agent's prediction accuracy is assessed per round. The experiments involve 3 types of simple

opponent models: (1) constant-action models, where the opponent repeatedly plays R, P, or S; (2) two-step loop models, such as R-P, P-S, and S-R; and (3) a three-step loop model cycling through R-P-S. The ability to recognize these behavioral patterns is evaluated by calculating the average number of points earned per round.

Neither `Llama3`, `DeepSeek-R1`, nor `Qwen3` successfully generated a valid prediction strategy. In contrast, `GPT-4.5` and `Mistral-Small` produced strategies that attempt to anticipate the opponent's next move by identifying the most frequently chosen action. These strategies prove effective against opponents with consistent behavior but break down when faced with more complex patterns. Figure 2a shows the average points earned per round (with 95% confidence intervals) for each LLM against the three constant-behavior opponents, under a one-shot action generation setting. Overall, model performance hovers near random, with the exceptions of `Mistral-Small` and `Qwen3`, which perform relatively better in this context. However, even these models struggle to handle opponents employing more sophisticated strategies.



(a) In the simplified RPS game

(b) In the RPS game

Fig. 2: Average points earned per round against constant behaviours

To assess the agents' ability to incorporate predictions of their opponent's next move into their decision-making, we analyze the performance of each generative agent in the Rock-Paper-Scissors (RPS) game. In this setup, a win awards 2 points, a draw 1 point, and a loss 0 points. Figure 2b shows the average points earned per round, along with 95% confidence intervals, for each LLM when facing constant-strategy opponents under a one-shot action generation setting. Although `Mistral-Small` and `Qwen3` can accurately predict their opponent's next move, they fail to incorporate this belief into their action selection.

In summary, generative autonomous agents struggle to anticipate and effectively respond to the actions of other agents in their decision-making processes.

6 Discussion

In this paper, we evaluated the capabilities of generative autonomous agents in simulating social decision-making processes. To this end, we focused on their ability to align with economic theory, account for preferences, make first- and second-order rational decisions, and anticipate the actions of other agents.

Our results show that GPT-4.5, Llama3, and Mistral-Small generally respect agent preferences but encounter greater difficulty in generating one-shot actions compared to producing strategies in algorithmic form. GPT-4.5 and Mistral-Small typically exhibit both first- and second-order economic and social rationality. In contrast, Llama3, despite often generating random strategies, adapts more effectively in one-shot settings—even though these decisions frequently diverge from principles of economic rationality. DeepSeek-R1, on the other hand, fails to develop valid strategies and performs poorly in generating actions consistent with preferences or rationality principles. Qwen3 aligns well with utilitarian preferences and moderately with altruistic ones but struggles to model egoistic and egalitarian preferences. It demonstrates strong first-order rationality when producing actions—particularly in guided or explicit contexts—but falls short in deeper inferential reasoning. Most critically, all evaluated LLMs struggle to anticipate the actions of other agents or to incorporate such beliefs effectively into their decision-making processes.

The key challenge for generative autonomous agents lies in refining their beliefs and integrating them effectively into decision-making to better adapt to their environment and interlocutors. This capacity is essential for fostering credible social behaviours, such as cooperation. In this respect, the use of the *Chain-of-Thought* (CoT) technique did not result in significant improvements. Nonetheless, other post-training techniques warrant further investigation, including test-time scaling methods to enhance reasoning during inference, and supervised fine-tuning approaches such as parameter-efficient fine-tuning (see [11] for a recent review).

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